CAMBRIDGE INTERNATIONAL EXAMINATIONS

Cambridge International General Certificate of Secondary Education

MARK SCHEME for the October/November 2014 series

0620 CHEMISTRY

0620/23 Paper 2 (Core Theory), maximum raw mark 80

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge will not enter into discussions about these mark schemes.

Cambridge is publishing the mark schemes for the October/November 2014 series for most Cambridge IGCSE®, Cambridge International A and AS Level components and some Cambridge O Level components.

® IGCSE is the registered trademark of Cambridge International Examinations.

1 **(a)** **(i)** B and D [2]

(ii) A [1]

(iii) C [1]

(iv) A [1]

(v) D [1]

(b) KBr [1]
allow: K^+Br

(c) 146 [2]
allow: 1 mark for correct atomic masses 19 and 32

[Total: 9]

2 **(a)** Any **four** from: [4]
solids: particles close together / no space between particles / particles arranged regularly / particles touching
solids: particles only vibrate
allow: particles cannot move / particles in fixed positions
liquids particles can slide over each other / particles have limited movement
ignore: particles can move unqualified
liquids: particles close together / particles not arranged regularly / particles arranged randomly / particles not in fixed positions
ignore: particles further apart than in solids
gases: particles far apart / particles arranged randomly
gases: particles can move everywhere / particles move anywhere / particles move randomly
note: It must be clear which state is being referred to
note: there must be reference to particles (or atoms / molecules / ions) in the answer to gain marks

(b) **(i)** A [1]

(ii) E and F [2]
allow: B

(iii) C and E [2]

(iv) B and F [2]

© Cambridge International Examinations 2014

(c) (i) 4th box down (last box) ticked [1]

(ii) argon is unreactive / inert [1]

air (or oxygen) may oxidise metals / air (or oxygen) may react with the (hot) metals / to prevent the air (or oxygen) reacting with the metals [1]

[Total: 14]

3 (a) (i) mortar [1]
allow: mortar and pestle

(ii) any suitable solvent other than water e.g. ethanol [1]
allow: ethanoic acid / aqueous ammonia
ignore: hydrochloric / sulfuric / nitric acids / strong alkalis / aqueous solutions of salts

(iii) evaporate some of the solvent [1]
allow: evaporate / heat
allow: add more rhubarb

(b) (i) it would dissolve / it would mix with the solvent / solvent would wash it off / so that the spot / Y didn't dissolve in the solvent / Z would dissolve in the solvent [1]

(ii) any **two** from: [2]

dip paper into the solvent
put lid on jar
let solvent run up paper / let solvent separate spots
ignore: wait for spots to appear / spots start to spread (unqualified)
take paper out before solvent reaches the top / record solvent front
ignore: reference to R_f values / locating agents

(c) (i) ring around one or both carboxylic acid groups; [1]
do not allow: ring around whole structure

(ii) $C_2H_2O_4$ [1]
ignore: $(COOH)_2$

© Cambridge International Examinations 2014

(d) **(i)** H_2O [1]

(ii) CO and CO_2 are gases / CO and CO_2 are given off / the products are gases (and water) [1]
ignore: other substances evaporated

(iii) any suitable source e.g. respiration / burning fuels / burning named carbon-containing fuel / from limekilns or other suitable decomposition reaction [1]
ignore: from burning (unqualified) / exhaled air / animals (unqualified)
allow: from car exhausts

(iv) any two of: [2]
it is a greenhouse gas / absorbs infrared radiation
allow: warms the atmosphere / traps heat in the atmosphere

causes global warming / increase temperature of the atmosphere
allow: warms the atmosphere / traps heat in the atmosphere
reject: absorbs heat from the Sun

effects of global warming e.g. desertification / rise in sea level / more extreme weather / climate change
ignore: references to ozone layer

[Total: 13]

4 **(a)** filter funnel with filter paper + container to collect filtrate [1]

correct labels for two of: (filter) funnel, filter paper, baker or flask [1]
ignore: incorrect labels
ignore: filtrate / water / sand

(b) **(i)** potassium nitrate [1]

(ii) Na^+ and CO_3^{2} (both required) [1]

(iii) sodium chloride [1]

(iv) total mass = 20 g [1]
% by mass = 14% [1]
allow: error carried forward from incorrect total mass

(c) **(i)** CO_2 [1]

(ii) pH 12 [1]

[Total: 9]

© Cambridge International Examinations 2014

5 **(a)** alcohol(s) / alkanol [1]

(b) O–H [1]
allow: OH

(c) **(i)** 3 (H_2) [1]

(ii) (hydrogen is) flammable / explosive [1]
allow: fire hazard

(CO is) poisonous / toxic [1]
ignore: CO harmful

(d) **(i)** decreases [1]
then remains constant [1]

(ii) 0.28 (mol / dm^3) [1]

(iii) **allow:** values between 44–46 (hours) [1]

(iv) curve steeper at start; [1]

curve levels out at same level and before 45 hrs [1]

(e) bonding pair of electrons between H and C*l* [1]
do not allow: if extra electrons on the H atom

Six non-bonding electrons around the C*l* [1]
ignore: inner shell electrons in C*l*

[Total: 13]

6 **(a)** **(i)** <u>acidic</u> oxide because oxide of non-metal [1]

(ii) Any **three** from: [3]
sulfur dioxide reacts with water in air / reacts with water on surface of building / forms acid rain
allow: sulfur dioxide is acidic / it is acidic
limestone is a carbonate
idea of reaction of acid with limestone / carbonate
carbon dioxide (+ salt + water) formed

© Cambridge International Examinations 2014

(b) **(i)** carry out in fume cupboard [1]

(ii) speeds up reaction [1]

(iii) O_2 (on left) [1]

correct balance (2 on right) [1]
note: second mark dependent on O_2 or 2O on left

(iv) to prevent it turning into liquid / vapour [1]
allow: so temperature is below melting point / so that it can form crystals

(v) 200 g [1]

(c) **(i)** 4th box down ticked (pipette) [1]

(ii) indication that indicator changes colour [1]
allow: any stated colour change

(d) water absorbed [1]

[Total: 13]

7 **(a)** Any **four** from: [4]
colour gets darker down the Group
correct colours of two of the halogens (chlorine green / yellow green + bromine brown / reddish-brown + iodine grey / grey-black / black)
note: all three halogen colours correct is 2 marks
correct state of two of the halogens (chlorine gas, bromine liquid, iodine solid)
note: all three states correct is 2 marks
reactivity decreases down the Group
allow: any two differences in reactivity correctly compared e.g. chlorine is more reactive than bromine (1 mark maximum)
do not allow: mention of incorrect difference in reactivity
example of reactivity of pair of halogens / halides e.g. chlorine reacts with potassium bromide
allow: density increases down Group
allow: boiling points / melting points get higher down the Group

(b) diatomic [1]

(c) 7 electrons in the outer shell [1]

2 electrons in inner shell [1]
note: this mark cannot be obtained if other inner shells are drawn

(d) bromine + potassium iodide → iodine + potassium bromide [2]

[Total: 9]

 © Cambridge International Examinations 2014